

Anamika Bharali

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WORK EXPERIENCE

Humanitarians AI | AI/ML - Data Scientist

Jan 2025 – Present

- Developed **multi-agent AI research platform** that automated market analysis to accelerate product validation
- Led **implementation & evaluation** of performance on integration of different LLMs/VLMs in multiple projects in PowerBI
- Developed AI agent with MCP client-server arch to recap meetings, check calendars, & coordinate org-wide scheduling

Veeco Instruments | Data Scientist Co-op

Jan 2024 – Jul 2024

- Saved \$100K yearly using CNN/ML models to predict silicon chip breakage and increased manufacturing accuracy by 94%
- Saved 200+ hours/\$90K annually by developing tool for automatic data analysis & processing all reports for cross-team use
- Engineered ETL pipelines & created dynamic **SQL-powered Tableau** dashboards to track machine health and detect process bottlenecks from automated Python scripts that read & clean machine logs for all product lines

Tata Motors | Senior Data Scientist

Jan 2021 – Jul 2022

- Increasing revenue by 19% utilizing **cluster analysis, market segmentation & hypothesis testing** to gain insights into high-value customer needs leading to successful integration of value-added services in Salesforce CRM
- Increased conversion rates by 15% with A/B testing, leading experiment design, & execution across full SDLC using JIRA
- Developed & deployed forecasting ML models in **AWS** with features designed with cross-functional collaboration with Finance and Product Teams, supporting pricing strategy and product planning
- Automated data ingestion and model retraining workflows using **Docker**, cutting manual intervention by 40%
- Developed comprehensive **PowerBI visualizations** to track product performance across daily, weekly, & MOM trends

Amazon | Data Scientist

Feb 2020 – Dec 2020

- Saved \$100K monthly costs by optimizing delivery algorithms with comprehensive **error analysis & A/B testing** in **BigQuery**
- Conducted **historical regression and time-series analysis** on month-wise data across several years, incorporating seasonal, weather, & local event trends to forecast operational needs, enabling effective staffing & resource planning
- Improved delivery efficiency by 35% with automated daily reports in **Quicksight and PowerBI** for all package routing KPIs
- Standardized complex SQL queries and **ETL** pipelines for last-mile delivery data extraction from Redshift databases

DRDO | Research Analyst Intern

May 2019 – Jul 2019

- Employed Maximum Likelihood & Method of moments techniques to derive precise predictions utilizing advanced statistical methodologies to forecast rainfall patterns, and conducted rigorous comparative analysis evaluating goodness fit

Indian Institute of Technology | Research Analyst Intern

May 2018 – Jul 2018

- Collaborated with Dr. Mayank Kumar, MIT Alum, and performed regression analysis and T-tests in Python on all existing flow metering techniques, providing valuable insights into their strengths and limitations for optimal oil production

PROJECTS

Integrated Financial Forecasting: AAPL Stock and Sentiment Analysis

Jul 2023

- Leveraged Alpha Vantage API to gather historical AAPL stock data and processed it with Pandas
- Developed 2 predictive models – one by integrating RNN & Transformer for one ensemble and another ensemble integrating Auto ARIMA, Facebook's Prophet, & a custom 2-layer neural network – and compared results in Streamlit

Enhanced Snowflake Database Exploration through Text-to-SQL: Livability Analysis

Oct 2023

- Created SQL/python processes & user-defined functions for Snowflake databases to analyze livability in 6 US cities
- Utilized Snowflake Snowpark to analyze integrated metrics & deployed on Google Cloud Platform
- Explored insights with visualizations, chatbot & text-to-SQL query features leveraging OpenAI, and Langchain in Streamlit

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, MS Excel DAX, Java

Tools and Libraries: Snowflake, Hadoop, MongoDB, GCP, AWS, Google Firestore, PySpark, Airflow, Pinecone Vector DB

Database & Visualization: SQL Server, PostgreSQL, MySQL, Azure Data Studio, MongoDB, Tableau, PowerBI, Alteryx, Seaborn

Statistical Techniques: Time-series analysis, Regression analysis, Descriptive statistics, Bayesian Statistics, Hypothesis testing

EDUCATION

Master of Science in Computer Software Engineering Northeastern University, Boston (MA)

2024

Bachelor of Engineering Thapar University, India

2020